

Material																				
Stream Name	Units	AIR	AIRSOURCE	COLD	AIR	COLD	EXH	COLD	TU	COM	GA	COMP	P	COMPR	COND	R	EVAPRE	EVAPRI	EXP	PREF FUEL
METHANOL WARM WATER																				
Description																				
From																				
To	MIX	EVAP1	EVAP1	EVAP2	TURBHE	REACT	C	FUEL	CC	PUMP	C	COND	EVAP1	EVAP2	REGEN	MIX	COND	EVAP	TURBHE	TURB
Stream Class	CONVEN	CONVEN	CONVEN	CONVEN	CONVEN	CONVEN	CONVEN	CONVEN	CONVEN	CONVEN	CONVEN	CONVEN	CONVEN	CONVEN	CONVEN	CONVEN	CONVEN	CONVEN	CONVEN	CONVEN
Maximum Relative Error	6.5E-06																			
MIXED Substream																				
Phase	Vapor	Phase	Vapor	Phase	Vapor	Phase	Vapor	Phase	Vapor	Phase	Vapor	Phase	Vapor	Phase	Vapor	Phase	Vapor	Phase	Vapor	Phase
Temperature	C	25	25	24.0525	25	232.91	1470.3	227.91	68.437	31	20	26.722	0.4671	24.9984	25	52.637	958.118	964.16	25	
Pressure	kPa	100	101.325	101.325	101.325	101.325	550	550	153.33	153.33	101.325	101.33	101.325	100	101.325	101.33	550	101.33	101.33	101.33
Molar Vapor Fraction	1	1	1	0.976718	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0
Molar Liquid Fraction	0	0	0	0.023282	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1
Molar Solid Fraction	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Mass Vapor Fraction	1	1	1	0.985312	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0
Mass Liquid Fraction	0	0	0	0.014688	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1
Mass Solid Fraction	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Molar Enthalpy	J/kmol	-6636.77791	-6724.61	-34416.7	-2.1E+07	-1E+07	3E+07	4E+06	1E+06	165373	-152847	43612	-723510	-1725084	-7.5E+07	-3E+08	2.8E+07	1E+07	-3E+08	
Mass Enthalpy	J/kg	-230.0411278	-233.085	-1192.94	-746897	-490301	995925	150893	43730	5732.1	-5297.9	1511.7	-25078	-60412.5	-4646133	-2E+07	995925	354738	-2E+07	
Molar Entropy	J/kmol-K	4361.663758	4251.945	4158.92	572.2076	19828.1	47323	4545.2	4779.1	1379.9	3757.69	4420.3	1743.15	3316.4	-80639.1	-156470	33909.1	49007	-163149	
Mass Entropy	J/kg-K	151.1821043	147.3791	144.155	20.03873	694.382	1657.3	159.17	165.65	47.831	130.247	153.21	60.4203	116.141	-5026.51	-8685.4	1187.5	1716.2	-9056.1	
Molar Density	kmol/cum	0.040346339	0.040881	0.04101	0.041864	0.02407	0.0379	0.1317	0.054	0.0606	0.04158	0.0406	0.04456	0.04035	0.04094	51.979	0.05365	0.0098	53.348	
Mass Density	kg/cum	1.164007896	1.179433	1.1832	1.19542	0.68741	1.0824	3.7611	1.5572	1.7495	1.19961	1.1726	1.28553	1.15213	0.65687	936.42	1.53203	0.2812	961.08	
Enthalpy Flow	Watt	-14.06022516	-11207.7	-57361.2	-46249.8	-30361	61670	9343.7	102037	13375	-12362	3527.2	-58515	-3740.9	-3726.84	-1E+07	61670.3	21966	-1E+07	
Average MW		28.8503972	28.8504	28.8504	28.55509	28.5551	28.555	28.555	28.85	28.85	28.8504	28.85	28.8504	28.5551	16.0428	18.015	28.5551	28.555	18.015	
Mole Flows	mol/min	127.111909	100000	100000	130.1119	130.112	130.11	130.11	4852.6	4852.6	4852.62	4852.6	4852.62	130.112	3	2220.3	130.112	130.11	2220.3	
WATER	mol/min	0	0	0	6	6	6	0	0	0	0	0	0	0	0	2220.3	0	6	2220.3	
OXYGE-01	mol/min	26.69350089	21000	21000	20.6935	20.6935	20.694	26.694	1019.1	1019.1	1019.05	1019.1	1019.05	26.6935	0	26.6935	20.694	0		
DIFLU-01	mol/min	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
CARBO-01	mol/min	0	0	0	3	3	3	0	0	0	0	0	0	0	0	0	0	3	0	
NITRO-01	mol/min	100.4184081	79000	79000	100.4184	100.418	100.42	100.42	3833.6	3833.6	3833.57	3833.6	3833.57	100.418	0	100.418	100.42	0		
METHA-01	mol/min	0	0	0	5.75E-32	5.7E-32	6E-32	3	0	0	0	0	0	3	3	0	3	6E-32	0	
Mole Fractions																				
WATER	0	0	0	0.046114	0.04611	0.0461	0	0	0	0	0	0	0	0	1	0	0.0461	1		
OXYGE-01	0.21	0.21	0.21	0.159044	0.15904	0.159	0.2052	0.21	0.21	0.21	0.21	0.21	0.21	0.20516	0	0	0.20516	0.159	0	
DIFLU-01	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
CARBO-01	0	0	0	0.023057	0.02306	0.0231	0	0	0	0	0	0	0	0	0	0	0	0.0231	0	
NITRO-01	0.79	0.79	0.79	0.771785	0.77178	0.7718	0.7718	0.79	0.79	0.79	0.79	0.79	0.79	0.77178	0	0	0.77178	0.7718	0	
METHA-01	0	0	0	4.42E-34	4.4E-34	4E-34	0.0231	0	0	0	0	0	0	0.02306	1	0	0.02306	4E-34	0	
Mass Flows	kg/min	3.667229063	2885.04	2885.04	3.715357	3.71536	3.7154	3.7154	140	140	140	140	140	3.71536	0.04813	40	3.71536	3.7154	40	
WATER	kg/min	0	0	0	0.108092	0.10809	0.1081	0	0	0	0	0	0	0	0	40	0	0.1081	40	
OXYGE-01	kg/min	0.854159996	671.9748	671.975	0.662167	0.66217	0.6622	0.8542	32.608	32.608	32.6084	32.608	32.6084	0.85416	0	0	0.85416	0.6622	0	
DIFLU-01	kg/min	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
CARBO-01	kg/min	0	0	0	0.132029	0.13203	0.132	0	0	0	0	0	0	0	0	0	0	0.132	0	
NITRO-01	kg/min	2.813069067	2213.065	2213.06	2.813069	2.81307	2.8131	2.8131	107.39	107.39	107.392	107.39	107.392	2.81307	0	0	2.81307	2.8131	0	
METHA-01	kg/min	0	0	0	9.22E-34	9.2E-34	9E-34	0.0481	0	0	0	0	0	0.04813	0.04813	0	0.04813	9E-34	0	
Mass Fractions																				
WATER	0	0	0	0.029093	0.02909	0.0291	0	0	0	0	0	0	0	0	1	0	0.0291	1		
OXYGE-01	0.232917001	0.232917	0.23292	0.178224	0.17822	0.1782	0.2299	0.2329	0.2329	0.23292	0.2329	0.23292	0.2299	0	0	0.2299	0.1782	0		
DIFLU-01	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
CARBO-01	0	0	0	0.035536	0.03554	0.0355	0	0	0	0	0	0	0	0	0	0	0	0.0355	0	
NITRO-01	0.767082999	0.767083	0.76708	0.757146	0.75715	0.7571	0.7571	0.7671	0.7671	0.76708	0.7671	0.76708	0.75715	0	0	0.75715	0.7571	0		
METHA-01	0	0	0	2.48E-34	2.5E-34	2E-34	0.013	0	0	0	0	0	0	0.01295	1	0	0.01295	2E-34	0	
Volume Flow	l/min	3150.519061	2446124	2438329	3107.993	5404.88	3432.4	987.84	89905	80022	116705	119389	108905	3224.77	73.2691	42.716	2425.12	13213	41.62	
Vapor Phase																				
Molar Enthalpy	J/kmol	-6636.77791	-6724.61	-34416.7	-1.5E+07	-1E+07	3E+07	4E+06	1E+06	165373	-152847	43612	-723511	-1725084	-7.5E+07		2.8E+07	1E+07		
Mass Enthalpy	J/kg	-230.0411278	-233.085	-1192.94	-519004	-490301	995925	150893	43730	5732.1	-5297.9	1511.7	-25078	-60412.5	-4646133		995932	354738		
Molar Entropy	J/kmol-K	4361.663758	4251.945	4158.92	4474.757	19828.1	47323	4545.2	4779.1	1379.9	3757.69	4420.3	1743.15	3316.4	-80639.1		33909.1	49007		
Mass Entropy	J/kg-K	151.1821043	147.3791	144.155	155.3394	694.382	1657.3	159.17	165.65	47.831	130.247	153.21	60.4203	116.141	-5026.51		1187.5	1716.2		
Molar Density	kmol/cum	0.040346339	0.040881	0.04101	0.04089	0.02407	0.0379	0.1317	0.054	0.0606	0.04158	0.0406	0.04456	0.04035	0.04094		0.05365	0.0098		
Mass Density	kg/cum	1.164007896	1.179433	1.1832	1.177883	0.68741	1.0824	3.7611	1.5572	1.7495	1.19961	1.1726	1.28553	1.15213	0.65687		1.53203	0.2812		
Enthalpy Flow	Watt	-14.06022516	-11207.7	-57361.2	-31666															

OXYGE-01	kg/min	0.854159996	671.9748	671.975	0.662167	0.66217	0.6622	0.8542	32.608	32.608	32.6084	32.608	32.6084	0.85416	0	0.85416	0.6622
DIFLU-01	kg/min	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
CARBO-01	kg/min	0	0	0	0.132029	0.13203	0.132	0	0	0	0	0	0	0	0	0	0.132
NITRO-01	kg/min	2.813069067	2213.065	2213.06	2.813069	2.81307	2.8131	2.8131	107.39	107.39	107.392	107.39	107.392	2.81307	0	2.81307	2.8131
METHA-01	kg/min	0	0	0	0	9.2E-34	9E-34	0.0481	0	0	0	0	0	0.04813	0.04813	0.04813	9E-34
Mass Fractions																	
WATER		0	0	0	0.01462	0.02909	0.0291	0	0	0	0	0	0	0	0	0	0.0291
OXYGE-01		0.232917001	0.232917	0.23292	0.180881	0.17822	0.1782	0.2299	0.2329	0.2329	0.23292	0.2329	0.23292	0.2299	0	0.2299	0.1782
DIFLU-01		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
CARBO-01		0	0	0	0.036066	0.03554	0.0355	0	0	0	0	0	0	0	0	0	0.0355
NITRO-01		0.767082999	0.767083	0.76708	0.768433	0.75715	0.7571	0.7571	0.7671	0.7671	0.76708	0.7671	0.76708	0.75715	0	0.75715	0.7571
METHA-01		0	0	0	0	2.5E-34	2E-34	0.013	0	0	0	0	0	0.01295	1	0.01295	2E-34
Volume Flow	l/min	3150.519061	2446124	2438329	3107.937	5404.88	3432.4	987.84	89905	80022	116705	119389	108905	3224.77	73.2691	2425.12	13213
Liquid Phase																	
Molar Enthalpy	J/kmol				-2.9E+08											-3E+08	-3E+08
Mass Enthalpy	J/kg				-1.6E+07											-2E+07	-2E+07
Molar Entropy	J/kmol-K				-163149											-156470	-163149
Mass Entropy	J/kg-K				-9056.11											-8685.4	-9056.1
Molar Density	kmol/cum				53.34775											51.979	53.348
Mass Density	kg/cum				961.0769											936.42	961.08
Enthalpy Flow	Watt				-14583.8											-1E+07	-1E+07
Average MW					18.01532											18.015	18.015
Mole Flows	mol/min				3.02922											2220.3	2220.3
WATER	mol/min				3.029214											2220.3	2220.3
OXYGE-01	mol/min				8.19E-07											0	0
DIFLU-01	mol/min				0											0	0
CARBO-01	mol/min				4.53E-06											0	0
NITRO-01	mol/min				1.56E-07											0	0
METHA-01	mol/min				0											0	0
Mole Fractions																	
WATER					0.999998											1	1
OXYGE-01					2.7E-07											0	0
DIFLU-01					0											0	0
CARBO-01					1.5E-06											0	0
NITRO-01					5.14E-08											0	0
METHA-01					0											0	0
Mass Flows	kg/min				0.054572											40	40
WATER	kg/min				0.054572											40	40
OXYGE-01	kg/min				2.62E-08											0	0
DIFLU-01	kg/min				0											0	0
CARBO-01	kg/min				1.99E-07											0	0
NITRO-01	kg/min				4.36E-09											0	0
METHA-01	kg/min				0											0	0
Mass Fractions																	
WATER					0.999996											1	1
OXYGE-01					4.8E-07											0	0
DIFLU-01					0											0	0
CARBO-01					3.65E-06											0	0
NITRO-01					8E-08											0	0
METHA-01					0											0	0
Volume Flow	l/min				0.056783											42.716	41.62

Work				
Stream Name	OPTWORK	REGWORK	TOTWORK	TURBWORK
POWER Watt	-26619.4103	-71890.2605	-98509.7	-39704
SPEED Hz				

Power				
Stream Name	OPTWORK	REGWORK	TOTWORK	TURBWORK
POWER Watt	-26619.4103	-71890.2605	-98509.7	-39704
SPEED Hz				